🛒 Procurement Plan — Rev 5.2 (Research Edition)

🛒 Cart 1: The “Mini-Project” Kit (Buy Now)

Objective:
Master the code stack, single-motor control, differential sensing, and the PID loop.

Estimated Total: $336.93

Item	Qty	Status	Product Name (Search Key)	Est. Price	Why
Brain	1	Teensy 4.1	Teensy 4.1 with Pins	$48.99	Main 600 MHz real-time controller
Wi-Fi	3	ESP32	ELEGOO ESP-WROOM-32	$16.99	Wireless telemetry bridge
Motor	3	NEMA 17	NEMA 17 Stepper 1.5A	$32.99	Test motor (becomes J4 later)
Driver	6	TMC2209	TMC2209 Stepper Driver	$29.99	Silent driver for development
Encoder	3	AS5600	AS5600 Encoder (White PCB)	$8.99	Absolute angle feedback (I^2C)
Noise Fix	10	Level Converter	HiLetgo 4 Channel IIC I2C Level Converter (Pack of 10)	$7.49	Boosts signal voltage (3.3V → 5V) to fight noise without expensive chips
Cables	1	Ethernet	Ethernet Cable (Cat5e)	$5.99	Carries differential I^2C signals

Item	Qty	Status	Product Name (Search Key)	Est. Price	Why
Power	1	Power Supply	Mean Well LRS-350-24	$28.54	24V, 15A (360W) PSU
Safety	1	Emergency Button	Red Mushroom E-Stop	$5.99	Emergency power cut-off
Resistance	1	Resistors	2.2kohm Resistor	$4.99	Noise Fix
Protection	1	Capacitor - May Have	100µF 35V Electrolytic Capacitor	$5.99	**CRITICAL**: Prevents TMC2209 failure
Noise	1	Ferrite Beads	Clip-On Ferrite Bead Kit	$9.99	Suppresses USB & power EMI
Proto	1	Already have	Breadboard Starter Kit	$0	Rapid wiring without soldering
Cables	2	Micro-USB -> USB-C Probably have	USB Cable Pack	$5.69 + 8.99	Micro-USB to USB-A(Teensy) & USB-C to USB-A (ESP32) **DATA CABLES**
Silicone hookup wire	1	—	BOJACK 16 AWG Silicone Wire Kit	$17.99	PSU → drivers (current, heat, vibration safe)
Silicone hookup wire	1	—	BOJACK 24 AWG Silicone Wire	$14.99	Steppers, enables, IO
Ferrules	1	—	SOMELINE Ferrule Crimping Kit	$21.99	Prevents screw terminals from loosening/arcing
JST-XH crimps + housings	1	—	XH 2.54mm Connector Kit	$13.49	Locking connectors for limits / encoders

Item	Qty	Status	Product Name (Search Key)	Est. Price	Why
Inline blade fuse + holder	1	—	Quelong Inline Fuse Holder	$3.98	PSU will not protect wiring or drivers
Common ground terminal block	1	—	MILAPEAK Screw Terminal Strip	$11.89	Star grounding; twisting fails
Limit switches	3–6	—	HiLetgo Micro Limit Switch KW12-3	$5.99	Homing impossible without them
Soldering Iron + solder	1	—	60W Temperature-Controlled Soldering Kit	$25	Crimps alone are fantasy; fixes inevitable

🛒 Cart 2: The “Heavy Metal” Kit (Buy Later)

Objective:
Build the physical robot and full mechanical structure.

Estimated Total: ~$378

> **Note:** Gears are listed as filament + grease since they are 3D printed.

Item	Qty	Status	Product Name (Search Key)	Est. Price	Why
Big Motors	2	—	NEMA 23 Stepper 3.0 Nm	$78	High torque for Base & Shoulder
Big Drivers	2	—	DM556T Stepper Driver	$38	Industrial drivers for NEMA 23
Small Motors	2	—	NEMA 11 Stepper Motor	$30	Wrist roll & gripper

Item	Qty	Status	Product Name (Search Key)	Est. Price	Why
Encoders	3	**UPDATED**	AS5600 Encoder (White PCB)	$8.99	Feedback for remaining joints
Noise Fix	8	**NEW**	PCA9615 Differential Breakout	$30	2 per joint (MCU + motor side)
Safety	1	**NEW**	24V 40A Automotive Relay	$5	Handles high-current E-Stop cut
Logic	1	**NEW**	4-Channel Level Shifter	$6	Required for full 6-axis system
Bearings	20	**NEW**	608ZZ Ball Bearings	$12	**CRITICAL**: Supports printed gears
Couplers	2	**NEW**	5mm → 8mm Rigid Coupler	$10	Connects NEMA 23 to gearbox
Gears	1	—	Black PETG Filament (1 kg)	$25	Printed gear material
Lube	1	—	White Lithium Grease	$8	Required for plastic gears
Hardware	1	—	M3 / M4 / M5 Screw Kit	$18	Mechanical assembly
Springs	1	—	Extension Spring Kit	$12	Gravity assist for Joint 3
Wiring	1	—	Shielded 4-Core Wire	$15	Stepper motor cabling

💰 Grand Total

Phase	Estimated Cost
Cart 1 — Electronics / Brain	~$213

Phase	Estimated Cost
Cart 2 — Mechanics / Structure	~$378
Total Project Cost	**~$591**

⚠️ Critical Build Rules

- **Capacitor Rule**
  Every **TMC2209** used on a breadboard **must** have a capacitor across **VM/GND**.
  Without it, the driver *will* fail on first deceleration.
- **Relay Rule**
  The **E-Stop button drives the relay coil**.
  The relay — *not the button* — carries the 15A motor current.
- **Ferrite Rule**
  Place ferrite beads on:
  - USB cable (Teensy end)
  - 24V power lines entering each driver